

Auditing Semantic and Structural Representations for Patent Outcome Prediction in Heterogeneous Temporal Networks

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Abstract. Patent intelligence systems increasingly rely on semantic, structural, and hybrid representations to score patents by technological, commercial, or international relevance. Such systems require temporally valid evaluation that separates relational signal from future-information leakage. We study patent outcome prediction, where patents are embedded in typed networks linking inventions, inventors, assignees, applications, examiners, and citations. We present a controlled, leakage-aware benchmark of structural, semantic, metadata, and fused representations on a large electric-vehicle patent knowledge graph built in Neo4j. Using a strict temporal split with patents from 2014–2016 for training and 2017 patents for testing, we compare Sentence-BERT text embeddings, node2vec embeddings, metadata features, and their combinations across renewal, triadic family membership, and citation-impact prediction. Random splitting inflates renewal AUC by 6.6 percentage points. Under temporal evaluation, structural embeddings outperform text for XGBoost across all tasks, including triadic prediction (0.749 vs. 0.726 AUC) and citation impact (0.576 vs. 0.433 Macro-F1), while metadata remains a strong low-cost baseline. The contribution is an empirical audit of representation quality under realistic temporal observability rather than a new embedding algorithm.

Keywords: heterogeneous networks · patent knowledge graphs · representation learning · temporal evaluation · leakage-aware benchmarking

1 Introduction

This paper studies how to audit representations for node-level prediction in temporal heterogeneous networks. Patent search and patent intelligence systems often score or rank patents by proxies for technological significance, commercial relevance, or international scope. Patents provide a useful testbed because they are embedded in relational systems connecting inventions to inventors, firms, examiners, applications, and citations. Outcomes such as renewal, triadic family membership, and citation impact therefore test whether a representation captures temporally observable relational structure rather than future leakage or task-specific artifacts.

Two representation paradigms are common in this setting. Semantic representations derived from transformer encoders capture patent titles and abstracts [9,2]. Structural representations learned from citation and collaboration networks encode a patent’s position in the innovation system [4,6]. Heterogeneous knowledge graphs (KGs) offer a natural way to integrate patents, inventors, assignees, applications, examiners, and typed edges in one relational system [8,11]. Recent patent studies have explored semantic, graph-based, and multimodal models, and some introduce temporal or leakage-aware elements [5,7,3,1]. However, the literature still lacks a compact controlled audit that isolates text, learned structure, simple network statistics, metadata, and fusion under the same temporally strict protocol.

Evaluation design is central. Random train-test splits allow future relational information to leak into training neighborhoods, inflating reported performance in temporal graphs [10]. In patent data this is especially problematic because future citations, renewal behavior, and international filing decisions are all correlated with latent patent quality. We therefore evaluate representations only from information observable at prediction time. Our setup is transductive in a limited, grant-time sense: 2017 test patents may be present as unlabeled nodes when embeddings are computed, but only with text, metadata, and edges observable on or before their grant dates. Future incoming citations, renewal outcomes, labels, and all 2018–2023 nodes and edges are hidden.

Our contribution is a leakage-aware empirical benchmark for representation auditing in heterogeneous temporal networks. We make three points. First, temporal observability matters: random splitting increases renewal AUC by 6.6 percentage points. Second, learned KG embeddings provide stronger structural signal than text-only embeddings and simple graph statistics in the primary XG-Boost benchmark. Third, metadata is a competitive low-dimensional baseline, and fusion helps most clearly for triadic prediction. Following the broader source manuscript, we also treat inventor-team composition as a fairness-sensitive control rather than as a neutral deployment feature. We deliberately use standard encoders and classifiers so that differences can be attributed to representation classes rather than to a new model architecture.

2 Data and Representations

2.1 Patent Knowledge Graph

We study 91,052 U.S. electric-vehicle patents granted between 2014 and 2023, identified through CPC-subclass filtering and keyword matching on titles and abstracts. The predictive cohort contains patents granted in 2014–2017. Later years are used only to compute forward-citation outcomes for the 2014–2017 cohort; they do not provide nodes, edges, or metadata features at prediction time.

The heterogeneous KG is constructed and managed in Neo4j. Nodes represent *Patent*, *Inventor*, *Assignee*, *Application*, and *Examiner* entities. Edges encode inventorship, assignment, citation, application, examination, continuation,

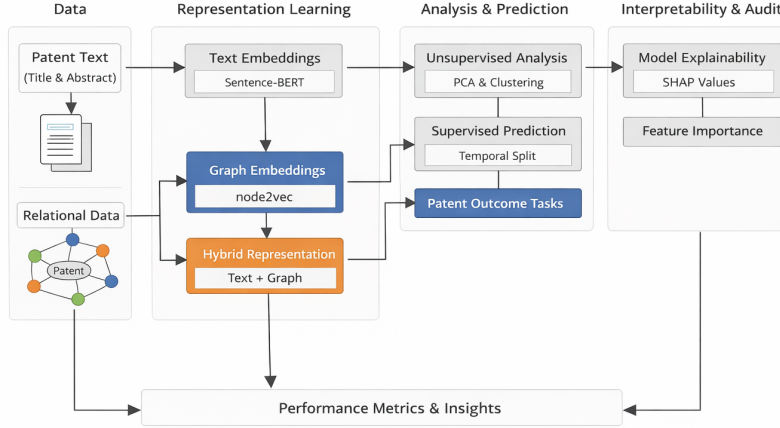


Fig. 1. Pipeline overview: Neo4j heterogeneous knowledge graph construction, temporal node2vec embedding, and multi-representation patent outcome prediction.

division, and reissue relationships. The graph contains more than 300K nodes and more than 700K edges. After validation, Neo4j exports node tables and edge lists for representation learning and downstream prediction.

We use a strict temporal split: 2014–2016 patents for training ($n = 20,469$) and 2017 patents for testing ($n = 8,915$). The three outcomes are: **4-year renewal** (`renewed_4y`), a binary maintenance-fee outcome with an 88.7% positive rate in the test set; **triadic family membership** (`is_triadic`), a binary outcome with a 25.0% positive test rate; and **citation impact**, a three-class discretization of forward citations into 0, 1–2, and 3+ citations with test shares of 50.7%, 30.1%, and 19.2%. AUC is primary for binary tasks and Macro-F1 for citation impact. The 2017 cohort also supports aggregate inventor-team composition checks: 78.6% male-only, 19.1% mixed-gender, and 2.4% female-only teams after unresolved cases are excluded.

2.2 Representation Families

TEXT uses all-mpnet-base-v2 Sentence-BERT embeddings of patent titles and abstracts, giving 768-dimensional L2-normalized vectors [9]. **KG** uses node2vec over the temporally masked heterogeneous graph, giving 64-dimensional patent embeddings [4]. **META** contains nine standardized grant-time or near-grant bibliographic attributes: number of claims, application processing days, priority count, backward citation count, international family size, grant lag, team size, female inventor count, and proportion of female inventors. **NETFEAT** contains six simple network counts: citation out-degree, inventor count, assignee count, family size, priority count, and team size. Citation in-degree is excluded because it directly leaks the citation target. We also evaluate simple concatenations: $\text{HYBRID} = \text{TEXT} + \text{KG}$, $\text{KG} + \text{META}$, and $\text{FULL} = \text{TEXT} + \text{KG} + \text{META}$.

Table 1. XGBoost performance under temporal evaluation (mean $\pm$ std, 10 seeds). AUC is primary for binary tasks; Macro-F1 is primary for citation impact.

Representation	Dim.	Renewal AUC	Triadic AUC	Citation Macro-F1
TEXT	768	0.588 $\pm$.005	0.726 $\pm$.002	0.433 $\pm$.001
KG	64	0.610 $\pm$.004	0.749 $\pm$.001	0.576 $\pm$.003
HYBRID	832	0.617 $\pm$.006	0.774 $\pm$.002	0.575 $\pm$.003
META	9	0.590 $\pm$.001	0.680 $\pm$.001	0.371 $\pm$.002
KG+META	73	0.645 $\pm$.003	0.783 $\pm$.002	0.580 $\pm$.002
FULL	841	0.647 $\pm$.004	0.795 $\pm$.002	0.578 $\pm$.002
NETFEAT	6	0.583 $\pm$.002	0.632 $\pm$.001	0.368 $\pm$.001

The gender-composition fields in META are treated as aggregate benchmark controls, not as recommended deployment features. They may encode sensitive demographic, institutional, and measurement structure, so we interpret them only as part of a low-cost metadata baseline and avoid causal claims about inventor gender.

3 Experimental Setup

We evaluate Logistic Regression (LR) and XGBoost (XGB) on the same temporal split. LR uses the `lbfgs` solver, $C = 1$, and `max_iter=500`. XGB uses 100 trees, maximum depth 6, learning rate 0.05, and the histogram method. Features are standardized using training-set statistics. Experiments are repeated over 10 random seeds. For XGB, seed variation reflects optimization variability on a fixed temporal split; it is not uncertainty over alternative temporal partitions. We therefore treat small differences, such as FULL vs. KG+META on renewal and citation impact, as descriptive. LR is deterministic under fixed features and serves as a qualitative robustness check rather than an inferential replicate.

4 Results

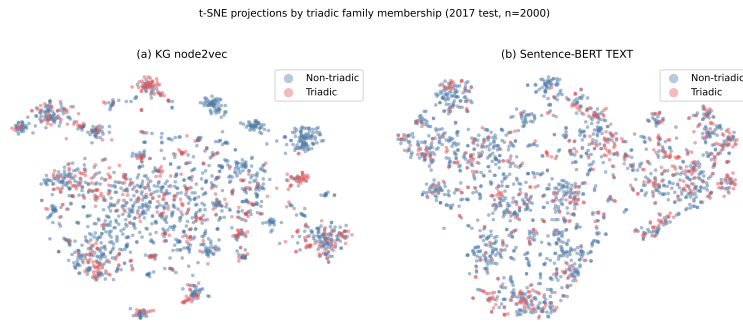
4.1 Main Representation Comparison

Table 1 reports the primary XGBoost results. KG embeddings outperform TEXT on all three tasks: renewal (+2.2 percentage points AUC), triadic membership (+2.3 points AUC), and citation impact (+14.3 points Macro-F1). The citation gap is the largest, suggesting that diffusion outcomes depend strongly on network position and institutional relationships that are weakly captured by text alone.

Fusion is most useful for binary outcomes. FULL reaches the best renewal and triadic scores, although the renewal gain over KG+META is only 0.2 percentage points. For citation impact, KG+META is descriptively highest and FULL is nearly identical. This pattern suggests that text adds complementary signal for triadic prediction but little additional information once structure and

Table 2. Logistic Regression robustness check (mean $\pm$ std, 10 seeds).

Representation	Renewal AUC	Triadic AUC	Citation Macro-F1
TEXT	0.557 $\pm$.000	0.693 $\pm$.000	0.430 $\pm$.000
KG	0.579 $\pm$.000	0.683 $\pm$.000	0.554 $\pm$.000
HYBRID	0.578 $\pm$.000	0.727 $\pm$.000	0.560 $\pm$.000
META	0.566 $\pm$.000	0.662 $\pm$.000	0.340 $\pm$.000
KG+META	0.610 $\pm$.000	0.706 $\pm$.000	0.566 $\pm$.000
FULL	0.596 $\pm$.000	0.736 $\pm$.000	0.569 $\pm$.000
NETFEAT	0.568 $\pm$.000	0.624 $\pm$.000	0.350 $\pm$.000

**Fig. 2.** t-SNE projections of KG node2vec (left) and Sentence-BERT TEXT (right) embeddings for 2017 test patents ($n = 2000$), colored by triadic family membership. KG space exhibits a denser triadic cluster, consistent with the XGBoost AUC gap.

metadata are already available for citation-impact classification. Metadata is also a strong low-cost baseline: KG+META approaches or matches FULL with only 73 dimensions rather than 841.

LR results preserve the broad renewal and citation pattern but partly reverse triadic ordering (Table 2): TEXT slightly exceeds KG for triadic prediction, and FULL exceeds KG+META. This model dependence supports a cautious interpretation. Relational structure is strongest in the primary nonlinear benchmark, but modality rankings can depend on classifier capacity and task geometry.

4.2 Ablations

Table 3 condenses the main ablations. First, random splitting inflates renewal prediction substantially: KG-XGB renewal AUC increases from 0.610 under temporal evaluation to 0.676 under a random split with the same sample sizes. Triadic inflation is smaller (+1.1 points), consistent with triadic status depending more on near-grant filing decisions than on post-grant citation dynamics.

Second, the KG advantage is not explained by simple graph statistics. The two degree-statistics baselines remain near chance, while node2vec over the same

Table 3. Ablation summary (XGB, 10 seeds). Values are AUC for renewal and triadic prediction.

Setting	Renewal	Triadic
KG, temporal split	$0.610 \pm .004$	$0.749 \pm .001$
KG, random split	$0.676 \pm .011$	$0.759 \pm .005$
Citation-graph statistics	$0.501 \pm .000$	$0.513 \pm .003$
Heterogeneous graph statistics	$0.518 \pm .000$	$0.499 \pm .004$
KG node2vec	$0.610 \pm .004$	$0.749 \pm .001$
NETFEAT only	$0.583 \pm .002$	$0.632 \pm .001$
KG + NETFEAT	$0.645 \pm .004$	$0.774 \pm .001$

Table 4. Cold/warm-start AUC by backward citation degree bucket. Patterns are clearest for triadic prediction; renewal is unstable in small high-degree buckets.

Task	Repr.	0 (6756)	1–2 (1614)	3–5 (341)	6–10 (114)	11+ (90)
Renewal	KG	0.608	0.631	0.543	0.804	0.138
	TEXT	0.580	0.613	0.494	0.786	0.345
	HYBRID	0.621	0.645	0.517	0.853	0.310
Triadic	KG	0.736	0.787	0.762	0.781	0.830
	TEXT	0.705	0.771	0.788	0.823	0.712
	HYBRID	0.759	0.821	0.813	0.849	0.773

temporally masked graph reaches 0.610 renewal AUC and 0.749 triadic AUC. The signal therefore lies in learned multi-hop embedding geometry rather than in raw degree counts alone. Third, NETFEAT is stronger than bare graph-statistics baselines because it includes bibliometric counts such as team size, family size, and priority count; nevertheless, KG embeddings outperform NETFEAT alone, and adding NETFEAT to KG improves both binary tasks.

An exploratory cold/warm-start analysis stratified 2017 patents by backward citation degree (Table 4). Triadic prediction showed a consistent HYBRID advantage across buckets, while renewal estimates in high-degree buckets were unstable because those buckets were small. We therefore treat the cold/warm-start pattern as suggestive and keep the main claims anchored in the full temporal test set.

5 Discussion

The results highlight three lessons for representation auditing in temporal heterogeneous networks. First, temporal evaluation is not a procedural detail. Random splits can expose future relational context and materially overstate predictive utility, especially for renewal. Second, learned structural representations are

more informative than simple graph counts. The near-chance degree-statistics baselines show that merely having graph data is not enough; the embedding procedure captures multi-hop relational context that raw statistics miss. Third, metadata deserves to be treated as a serious baseline. In this study, nine bibliographic features often approach high-dimensional fused representations, which matters for settings where text or graph embeddings are expensive to compute or difficult to deploy.

The citation-impact task remains the hardest. Macro-F1 values are modest even for the best models, and the fixed-threshold discretization of forward citations loses count-level detail. A plausible mechanism is that citation diffusion depends on network visibility, institutional position, and later technological uptake, while title and abstract text mostly capture technical content at grant time. This helps explain why KG outperforms TEXT by a much larger margin for citation impact than for renewal or triadic membership.

The broader patent-intelligence source manuscript also motivates a fairness caution: when model scores are used for ranking, representation choice can shift which inventor-team groups receive top- k visibility. This short ASONAM version does not claim that any representation is exposure-neutral. Systems that deploy these scores should audit group exposure, report no-gender sensitivity checks, and avoid using inferred gender composition as an operational ranking signal.

The study has limitations. It covers U.S. EV patents and one primary temporal split, so generality across domains, patent offices, and longer time periods remains open. Node2vec is static and does not model directed citation semantics or evolving temporal interactions. Inventor-team gender composition is inferred and may contain measurement error, especially across naming conventions. We use simple concatenation for fusion to preserve interpretability, but learned or attention-weighted fusion may exploit modality complementarity more effectively. Future work should evaluate dynamic graph encoders, heterogeneous GNNs, patent-specialized text models, and multiple temporal/domain splits.

6 Conclusion

We presented a leakage-aware empirical audit of semantic, structural, metadata, and fused representations for patent outcome prediction in a temporal heterogeneous network. Using a Neo4j EV patent KG and a 2014–2016 to 2017 temporal split, we showed that random splitting inflates renewal AUC, that learned KG embeddings outperform text-only embeddings in the primary XG-Boost benchmark, and that metadata is a strong low-cost baseline. The broader message is that conclusions about representation quality and patent-intelligence ranking can change once temporal observability and fairness-sensitive metadata are handled carefully. Open code and data artifacts are available at <https://zenodo.org/records/18665147>.

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